

Ex.No. 15: AI-based Traffic Prediction for 5G Networks

Objective1:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time samples
```

```
t = 1:10;
```

```
% Prediction Accuracy (%)
```

```
Accuracy = [90 91 92 93 94 95 96 97 98 99];
```

```
% Actual Network Traffic (Mbps)
```

```
Traffic = [45 50 55 60 65 70 75 80 85 90];
```

```
figure
```

```
...xlabel('Time Interval')
```

```
ylabel('Prediction Accuracy (%)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
bar(Traffic)
```

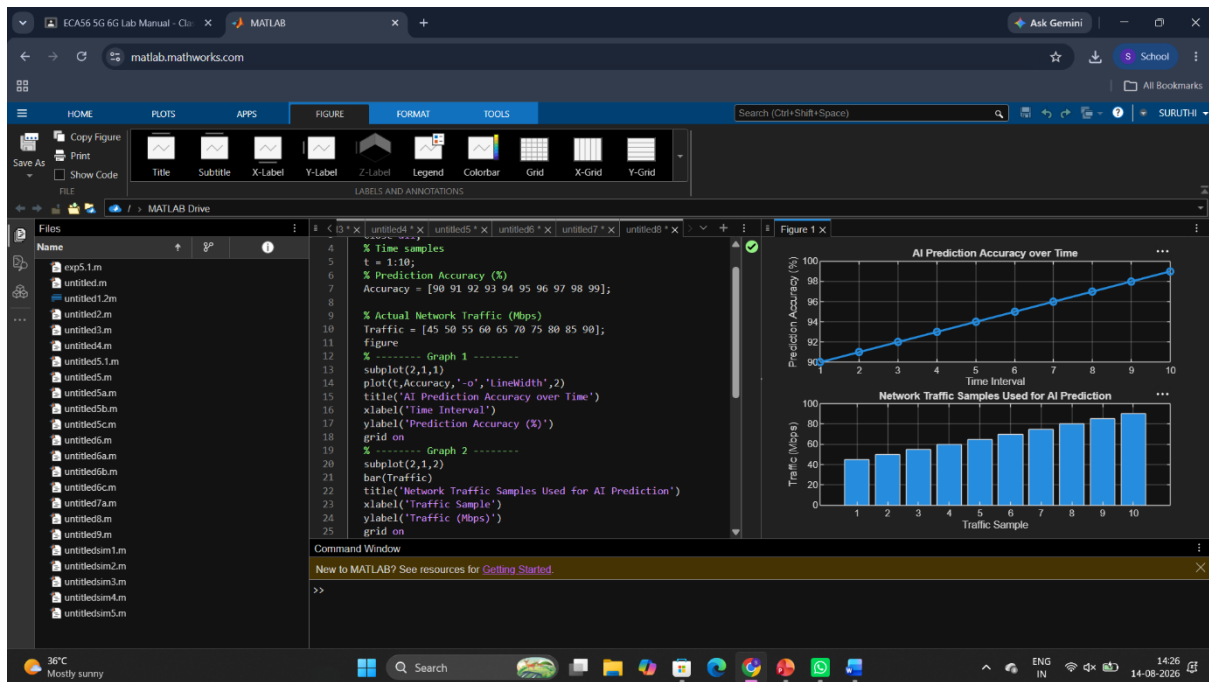
```
title('Network Traffic Samples Used for AI Prediction')
```

```
xlabel('Traffic Sample')
```

```
ylabel('Traffic (Mbps)')
```

```
grid on
```

output:



Objective2:

Program:

clc;

clear;

close all;

% Time intervals

t = 1:10;

% Actual and Predicted Traffic (Mbps)

Actual = [50 55 60 62 68 70 75 80 85 90];

Predicted = [49 54 61 63 67 71 74 81 84 89];

% Squared Error

SqError = (Actual - Predicted).^2;

...xlabel('Time Interval')

ylabel('Squared Error')

grid on

% ----- Graph 2 -----

```
subplot(2,1,2)
```

```
bar([MSE])
```

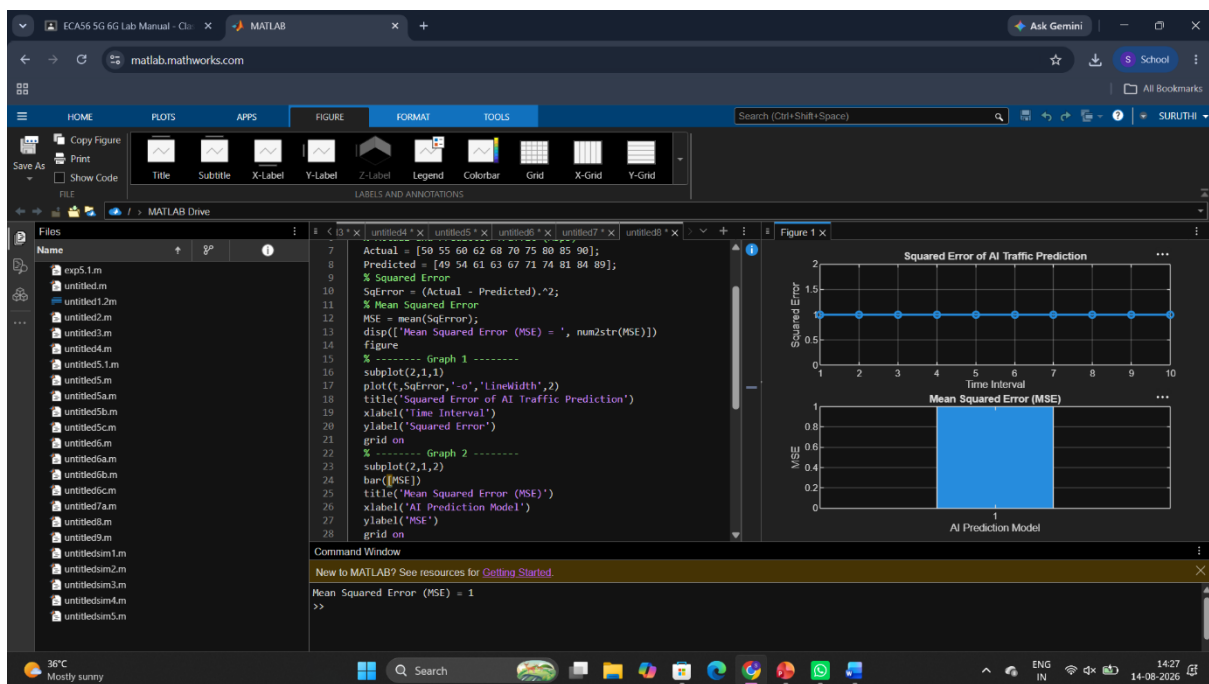
```
title('Mean Squared Error (MSE)')
```

```
xlabel('AI Prediction Model')
```

```
ylabel('MSE')
```

```
grid on
```

output:



Objective3:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time intervals
```

```
t = 1:10;
```

```
% Network Throughput (Mbps)
```

```
Throughput = [120 135 150 165 180 195 210 225 240 255];
```

```
% Network Utilization (%)
```

```
Utilization = [60 65 70 74 78 82 86 90 94 98];
```

figure

```
...% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(Throughput,Utilization,'-s','LineWidth',2)
```

```
title('Throughput vs Network Utilization')
```

```
xlabel('Network Throughput (Mbps)')
```

```
ylabel('Network Utilization (%)')
```

```
grid on
```

```
disp('Time Throughput(Mbps) Utilization(%))')
```

```
disp([t' Throughput' Utilization'])
```

output:

